Table of contents

Questions

1. What is a Gaussian Mixture model (GMM)?

Intuition:

A Gaussian Mixture Model (GMM) is a probabilistic model that represents a complex data distribution as a weighted sum of multiple Gaussian (normal) distributions. Instead of assuming data comes from a single bell curve, it models data as coming from several overlapping bell curves.

Detailed Answer:

Formally, a GMM is defined as:

$$f(x) = \sum_{j=1}^{c} \pi_j \phi(x, \theta_j)$$

where:

- c is the number of Gaussian components (clusters)
- π_j are the mixture weights with $\sum_{j=1}^{c} \pi_j = 1$ and $\pi_j \geq 0$
- $\phi(x, \theta_j)$ is the Gaussian density function for component j:

$$\phi(x, \theta_j) = \mathcal{N}(x|\mu_j, \Sigma_j) = \frac{1}{(2\pi)^{D/2}|\Sigma_j|^{1/2}} \exp\left(-\frac{1}{2}(x-\mu_j)^T \Sigma_j^{-1}(x-\mu_j)\right)$$

The parameters to estimate are $\theta = \{\pi_j, \mu_j, \Sigma_j\}_{j=1}^{c}$. GMMs are universal approximators of smooth densities and are particularly useful for modeling multi-modal distributions.

---

2. Why can it be viewed as conditional modeling?

Intuition:

GMM can be interpreted as a generative process where each data point is produced by first randomly selecting one of the Gaussian components according to the mixture weights, and then sampling from that selected component.

Detailed Answer:

We can introduce a latent (hidden) categorical random variable $z \in \{1, \dots, c\}$ that indicates which component generated a data point. The generative process is:

1. Sample $z \sim \text{Categorical}(\pi_1, \dots, \pi_c)$
2. Given $z = j$, sample $x \sim \mathcal{N}(\mu_j, \Sigma_j)$

Thus, the joint distribution factorizes as:

$$p(x, z) = p(z)p(x|z)$$

where:

- $p(z = j) = \pi_j$
- $p(x|z = j) = \mathcal{N}(x|\mu_j, \Sigma_j)$

The marginal distribution of x is obtained by summing over z:

$$p(x) = \sum_{j=1}^{c} p(z = j)p(x|z = j) = \sum_{j=1}^{c} \pi_j \mathcal{N}(x|\mu_j, \Sigma_j)$$

This is exactly the GMM formulation, making it a conditional model where x is conditioned on the latent variable z.

---

3. How can we use it for generating data?

Intuition:

To generate new data from a trained GMM, we simulate the generative process: randomly pick a component according to the mixture weights, then draw a sample from the selected Gaussian distribution.

Detailed Answer:

Given estimated parameters $\{\pi_j, \mu_j, \Sigma_j\}_{j=1}^c$, the data generation procedure is:

1. **Sample component index:** Randomly choose a component j with probability π_j.
2. **Sample from Gaussian:** Generate a vector x from the multivariate normal distribution $\mathcal{N}(\mu_j, \Sigma_j)$.

Mathematically:

- Draw $j \sim \text{Categorical}(\pi_1, \dots, \pi_c)$
- Draw $x \sim \mathcal{N}(\mu_j, \Sigma_j)$

This process can be repeated to generate any number of synthetic data points that follow the same distribution as the training data. This is useful for data augmentation, Monte Carlo simulations, or as a component in more complex generative models.

---

4. What is the responsibility? How to interpret it?

Intuition:

Responsibility γ_{ij} is a "soft" assignment weight that represents the probability or degree to which data point i belongs to component j, given the current model parameters.

Detailed Answer:

The responsibility γ_{ij} is defined as:

$$\gamma_{ij} = p(z_j = 1 | x_i, \theta) = \frac{\pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j)}{\sum_{k=1}^c \pi_k \mathcal{N}(x_i | \mu_k, \Sigma_k)}$$

This is obtained using Bayes' theorem:

- π_j is the prior probability of component j
- $\mathcal{N}(x_i | \mu_j, \Sigma_j)$ is the likelihood of x_i under component j
- The denominator is the marginal likelihood (evidence)

Interpretation:

- $\gamma_{ij} \in [0, 1]$
- $\sum_{j=1}^{c} \gamma_{ij} = 1$ for each data point i
- γ_{ij} represents the posterior probability that component j generated data point i
- It quantifies the "responsibility" that component j takes for explaining x_i

In the EM algorithm, responsibilities are computed in the E-step and used as soft assignments in the M-step to update the parameters.

---

5. Why is EM for GMM a MLE?

Intuition:

The EM algorithm for GMM is a method to find the parameters that maximize the likelihood of the observed data, making it a Maximum Likelihood Estimation (MLE) approach.

Detailed Answer:

Given data $X = \{x_1, \ldots, x_N\}$, the log-likelihood under a GMM is:

$$LL(\theta; X) = \sum_{i=1}^{N} \log \left(\sum_{j=1}^{c} \pi_j \mathcal{N}(x_i | \mu_j, \Sigma_j) \right)$$

Direct maximization is difficult due to the sum inside the logarithm. EM addresses this by:

1. **E-step:** Compute responsibilities γ_{ij} (expected values of latent indicators)
2. **M-step:** Maximize the expected complete-data log-likelihood:

$$Q(\theta, \theta^{\text{old}}) = \sum_{i=1}^{N} \sum_{j=1}^{c} \gamma_{ij} \left[\log \pi_j + \log \mathcal{N}(x_i | \mu_j, \Sigma_j) \right]$$

The M-step updates $(\pi_j^{\text{new}}, \mu_j^{\text{new}}, \Sigma_j^{\text{new}})$ are obtained by setting derivatives of Q to zero, yielding:

$$\pi_j^{\text{new}} = \frac{\sum_i \gamma_{ij}}{N}, \quad \mu_j^{\text{new}} = \frac{\sum_i \gamma_{ij} x_i}{\sum_i \gamma_{ij}}, \quad \Sigma_j^{\text{new}} = \frac{\sum_i \gamma_{ij} (x_i - \mu_j^{\text{new}})(x_i - \mu_j^{\text{new}})^T}{\sum_i \gamma_{ij}}$$

EM guarantees that $LL(\theta^{\text{new}}; X) \geq LL(\theta^{\text{old}}; X)$, iteratively increasing the likelihood until convergence to a (local) maximum. Thus, EM for GMM performs MLE.

---

6. How to interpret the E-step?

Intuition:

The E-step (Expectation step) computes the expected values of the latent variables (component assignments) given the current parameters and observed data. It "fills in" the missing information needed for the M-step.

Detailed Answer:

In the E-step, we compute the responsibilities γ_{ij} for all i, j:

$$\gamma_{ij} = \mathbb{E}[z_{ij}|x_i, \theta^{\text{old}}] = p(z_j = 1|x_i, \theta^{\text{old}})$$

where z_{ij} is an indicator variable: $z_{ij} = 1$ if x_i belongs to component j, else 0.

Interpretation:

- We are computing the posterior probability of each latent assignment
- This is a "soft" assignment, as opposed to hard assignment (like in k-means)
- The E-step uses Bayes' theorem to update our beliefs about which component generated each data point, based on current parameter estimates
- These responsibilities are then used as weights in the M-step to update the parameters

The E-step essentially completes the data by providing expected values for the missing latent variables, enabling the subsequent maximization in the M-step.

---

7. How to interpret the M-step?

Intuition:

The M-step (Maximization step) updates the model parameters by maximizing the expected complete-data log-likelihood, using the responsibilities computed in the E-step as weights.

Detailed Answer:

Given responsibilities γ_{ij} from the E-step, the M-step updates the parameters $\theta = \{\pi_j, \mu_j, \Sigma_j\}$ by maximizing:

$$Q(\theta, \theta^{\text{old}}) = \sum_{i=1}^{N} \sum_{j=1}^{c} \gamma_{ij} \left[\log \pi_j + \log \mathcal{N}(x_i|\mu_j, \Sigma_j)\right]$$

Interpretation:

- The function Q is the expected complete-data log-likelihood, where the expectation is taken with respect to the posterior distribution of z computed in the E-step
- The update formulas are weighted versions of the standard MLE for Gaussian distributions:
  - π_j is updated to the average responsibility of component j
  - μ_j is updated to the weighted mean of data points, with weights γ_{ij}
  - Σ_j is updated to the weighted covariance matrix
- The weights γ_{ij} ensure that data points contribute more to parameters of components they are likely to belong to

The M-step guarantees that the log-likelihood of the observed data does not decrease (and typically increases) with the new parameters.

---

8. Discuss the relationship between GMM and k-means

Intuition:

k-means can be seen as a special, simplified case of GMM where cluster assignments are "hard" (each point belongs to exactly one cluster) and clusters are spherical with equal variance.

Detailed Answer:

Similarities:

- Both are unsupervised clustering methods
- Both require specifying the number of clusters/components c
- Both are sensitive to initialization

Differences:

Aspect	k-means	GMM (EM)
Assignment	Hard: each point assigned to one cluster	Soft: each point has probabilities of belonging to each component
Cluster shape	Spherical (based on Euclidean distance)	Ellipsoidal (covariance matrices can be full, diagonal, or spherical)

Aspect	k-means	GMM (EM)
Probabilistic	No	Yes, provides likelihoods
Algorithm	Minimizes sum of squared distances	Maximizes log-likelihood via EM
Convergence	May converge faster	Slower due to soft assignments

Relationship: k-means is a special case of GMM when:

- All covariance matrices are spherical and identical: $\Sigma_j = \sigma^2 I$ for all j
- The mixing weights are equal: $\pi_j = 1/c$
- The variance $\sigma^2 \to 0$
- In this limit, responsibilities become binary (hard assignments), and the EM algorithm reduces to a form similar to k-means.

Thus, GMM is a more flexible, probabilistic generalization of k-means.

---

9. What are hard- and soft-assignments?

Intuition:

Hard assignment means each data point belongs exclusively to one cluster. Soft assignment means each data point has a probability distribution over clusters, representing degrees of membership.

Detailed Answer:

Hard Assignment:

- Each data point x_i is assigned to exactly one cluster j^*
- $j^* = \arg\max_j \gamma_{ij}$ (in the context of GMM)
- Binary membership: $\gamma_{ij} \in \{0, 1\}$
- Used in algorithms like k-means
- Advantages: Simpler, faster computation
- Disadvantages: May not capture uncertainty or overlapping clusters

Soft Assignment:

- Each data point x_i has a vector of responsibilities $(\gamma_{i1}, \dots, \gamma_{ic})$
- $\gamma_{ij} \in [0, 1]$ and $\sum_j \gamma_{ij} = 1$
- Represents posterior probabilities of cluster membership
- Used in EM for GMM

- Advantages: More expressive, handles uncertainty, allows for overlapping clusters
- Disadvantages: More computationally intensive, may converge slower

In practice, soft assignments are more informative but require more computation. Many algorithms can be adapted to use either approach.
